AFC Transaction Monitoring API — Integration Guide

Nov 3, 2026 · @Mashrukh

What the service does

Every night the service scans the last 24 hours of transactions in MongoDB and flags risky ones. It looks at each transaction individually and in context: velocity, structuring, fan-in and fan-out, and pass-through.

Three independent detectors score each transaction:

Detector	Method	Evidence returned
`rule`	Monitoring rules from the Rule Engine, plus collective typology rules	Matched conditions with expected and observed values
`ml`	LightGBM model	Top contributing fields (TreeSHAP)
`dl`	Neural network (PyTorch)	Top contributing fields (Integrated Gradients)

Each risky transaction produces three things:

- A record in `transaction-monitoring-ai-results`, which is what this API serves
- One `Alert` with `alertOrigin: "AI Based"`
- A link to a `Case` that groups similar alerts for the same customer and case type

Models retrain daily at 00:15 and monitoring runs at 01:00 (Asia/Dhaka). Customer names and account numbers are masked in all responses.

Access

- **Base URL:** `http://[TM_HOST]:5054/api/v1/tm` (ask the Data & AI team for the host)
- **Auth:** send `X-API-Key: [YOUR_API_KEY]` on every request. A missing or wrong key returns `401`.
- **Format:** JSON. IDs are 24-hex ObjectId strings and timestamps are ISO-8601 UTC.
- **Dates:** query parameters use `YYYY-MM-DD` in Asia/Dhaka time.
- **Errors:** `401` auth, `404` not found, `422` invalid parameter (with a `detail` message).
- **Health:** `GET http://[TM_HOST]:5054/health` (no key needed)

Keep API keys server-side and never in front-end code. Responses contain transaction-level AML data.

Endpoints

Method	Path	Returns
GET	`/results/daily?date=2026-11-03`	Risky transactions for one day, plus a `summary` for dashboards

Method	Path	Returns
GET	`/results?start=2026-10-01&end=2026-10-31`	Risky transactions in a date range (max 366 days); add `includeSummary=true` for the summary
GET	`/results/timeseries?start=…&end=…`	Per-day counts by detector, risk label and case type, with average score
GET	`/results/{transactionId or uid}`	Full detection record for one transaction
GET	`/runs?jobType=monitoring`	Recent pipeline runs with counts and model versions
GET	`/models`	Model versions and their metrics
GET	`/rules/coverage`	Which rules ran in the latest run, and why the others were skipped
POST	`/runs/monitoring`	Start a monitoring run (`202` with `runId`); for operators

Filters (on the list endpoints):

- `detector`: `rule`, `ml` or `dl`
- `riskLabel`: `Low`, `Medium`, `High` or `Critical`
- `caseType`: `Fraud`, `AML`, `Compliance` or `TF`
- `ruleId` (for example `R5`) and `field` (a triggered field, for example `amount`)
- `clientId`, `customerId` and `minScore` (0–100)
- `dateField`: `run` (default, the date the run happened) or `transaction` (the date the transaction happened)

Paging and shape: `page`, `limit` (max 500), `sort` (`score`, `time` or `detected`), and `view`. `view=compact` (the default) omits matched-condition details; `view=full` includes them.

The `summary` object contains:

- Totals, and amount by currency
- Counts by detector, by detector combination, by risk label, by case type and by domain
- Top rules and top triggered fields
- Score bands and alert outcomes

Result record

There is one record per risky transaction. Re-running a day updates the existing record instead of adding a new one.

Field	Meaning
`transactionId`, `transactionUid`, `customer`, `client`	References for joining back to source data

Field	Meaning
`runDate`, `transactionDate`, `detectedAt`	When the transaction was detected and when it happened
`transaction`	Amount, currency, channel, type, cross-border flag, and masked sender and beneficiary
`overall`	`riskScore` (0–100), `riskLabel`, `priority`, `detectedBy[]`, `agreement` (e.g. `2/3`), `primaryCaseType`, `primaryReason`
`ruleHits[]`	Rules that fired, each with `matchedConditions` (field, operator, expected, actual)
`modelDetections.ml` / `.dl`	Probabilities, threshold, and `topFactors[]` (field, value, contribution)
`triggeredFields[]`	One entry per field across all detectors, with `weight`, `detectors[]` and `value`. This is the main input for "why flagged" charts.
`reasons[]`	One readable sentence per detection
`categories`	`caseTypes`, `domains`, `mainDomains` for grouping
`collectiveContext`	24-hour counts and sums, unique counterparties, 30-day z-score, frequency change, pass-through ratio
`alert`, `case`	IDs and uids of the alert and case that were created

Sample responses

These responses were captured from the service's own endpoints, run on seeded demo data. Long arrays are cut to one to three entries, and IDs are demo values.

GET /results/daily?date=2026-11-03&limit=1

Each item is abridged here to its key fields. The full item has the same structure as the single-transaction record below, but without matched-condition details (default `view=compact`). `GET /results` (date range) returns the same shape, with `start` and `end` instead of `date`.


```
{
  "date": "2026-11-03",
  "dateField": "run",
  "pagination": {
    "page": 1,
    "limit": 1,
    "total": 106,
    "pages": 106
```

```
  },
  "items": [
    {
      "transactionId": "6ae9c7b4a469d23ffb7ea51b",
      "transactionUid": "TXN_1793640900000109",
      "customer": "6ae9c7b3a469d23ffb7e92c6",
      "runDate": "2026-11-03",
      "transactionDate": "2026-11-02",
      "transactionTimestamp": "2026-11-02T17:35:00Z",
      "overall": {
        "isRisky": true,
        "riskScore": 100,
        "riskLabel": "High",
        "priority": "high",
        "detectedBy": ["rule", "ml", "dl"],
        "detectorCount": 3,
        "agreement": "3/3",
        "primaryCaseType": "Fraud",
        "primaryReason": "Rule R84 burst_large [Fraud/High]: Sum of last 5
transactions gt 1000000 (observed 2253360.59)"
      },
      "topTriggeredFields": [
        "derived.sender_sum_last_5",
        "derived.amount_pct_decrease_prev",
        "derived.agg_24h_txn_value",
        "beneficiary.institution"
      ],
      "alert": {
        "alertId": "6ae9c802a469d23ffb7ec059",
        "uid": "AL-0000133",
        "status": "new",
        "action": "created"
      },
      "case": {
        "caseId": "6ae9c802a469d23ffb7ec0bf",
        "uid": "CA-1793705986416-261665",
        "action": "created"
      }
    }
  ],
  "summary": {
    "total": 106,
    "amountByCurrency": [
      { "currency": "BDT", "sum": 9760976.56, "count": 105 },
      { "currency": "USD", "sum": 891006.99, "count": 1 }
    ],
    "byDetector": { "dl": 89, "ml": 96, "rule": 102 },
    "byDetectorCombination": { "rule+ml+dl": 80, "rule+ml": 16, "rule+dl": 5, "dl":
4, "rule": 1 },
    "byRiskLabel": { "Critical": 1, "High": 101, "Medium": 4 },
    "byCaseType": { "AML": 4, "Fraud": 102 },
    "byDomain": {
      "Velocity & Pattern Shift": 99,
      "Transaction Amount": 77,
      "Account & Customer Activity Rules": 33
```

```
    },
    "topRules": [
      { "ruleId": "R82", "ruleName": "sudden_amt_increase", "caseType": "Fraud",
"count": 42 },
      { "ruleId": "R83", "ruleName": "sudden_amt_drop", "caseType": "Fraud", "count":
38 },
      { "ruleId": "R223", "ruleName": "sudden_freq_increase", "caseType": "Fraud",
"count": 29 }
    ],
    "topFields": [
      { "field": "derived.pass_through_ratio", "count": 79, "totalWeight": 5.2469 },
      { "field": "beneficiary.institution", "count": 78, "totalWeight": 7.0506 },
      { "field": "amount", "count": 76, "totalWeight": 46.6941 }
    ],
    "scoreBands": { "40-69": 2, "70-89": 5, "90-100": 99 },
    "alerts": { "created": 106 }
  }
}
```


GET /results/{transactionId}

This returns the full record, including `matchedConditions`, `triggeredFields[].sources`, `mitigatingFactors` and `runHistory`. The same record is returned by `uid`, for example `/results/TXN_1793628900000246`.


```
{
  "transactionId": "6ae9c7b3a469d23ffb7e9604",
  "_id": "6ae9c802a469d23ffb7ebfd0",
  "transactionUid": "TXN_1793628900000246",
  "client": "6ae9c7b3a469d23ffb7e8f31",
  "branch": "6ae9c7b3a469d23ffb7e8f32",
  "customer": "6ae9c7b3a469d23ffb7e935f",
  "counterpartyCustomer": null,
  "customerUid": "CR_1700000000156",
  "subjectKey": "cust:6ae9c7b3a469d23ffb7e935f",
  "runId": "MON-20261102T180000Z-828c0c",
  "runDate": "2026-11-03",
  "transactionDate": "2026-11-02",
  "transactionTimestamp": "2026-11-02T14:15:00Z",
  "detectedAt": "2026-11-03T11:39:46.082000Z",
  "window": {"start": "2026-11-01T18:00:00Z", "end": "2026-11-02T18:00:00Z"},
  "transaction": {
    "type": "transfer",
    "subtype": "",
    "amount": 27840.7,
    "currency": "BDT",
    "convertedAmountAUD": 356.93,
    "channel": "Mobile",
    "status": "completed",
    "purpose": "",
    "isCrossBorder": 0.0,
    "sender": {
      "customer": "6ae9c7b3a469d23ffb7e935f",
      "name": "C*** 0***",
      "account": "****0156",
```

```
      "institution": "Eastern Bank",
      "country": "Bangladesh",
      "bic": null
    },
    "beneficiary": {
      "customer": null,
      "name": "P*** 3***",
      "account": "****0657",
      "institution": "BRAC Bank",
      "country": "Bangladesh",
      "bic": null
    },
    "beneficiaryFromReceiver": false,
    "crypto": {"network": null, "cluster": null, "hops": null, "hasWallet": false}
  },
  "overall": {
    "isRisky": true,
    "riskScore": 100,
    "riskLabel": "High",
    "priority": "high",
    "detectedBy": ["rule", "ml", "dl"],
    "detectorCount": 3,
    "agreement": "3/3",
    "primaryCaseType": "Fraud",
    "primaryReason": "Rule R121 unknown_sof [Fraud/High]:
customer.personalKyc.funds_wealth.source_of_funds eq UNKNOWN (observed UNKNOWN)"
  },
  "ruleHits": [
    {
      "ruleRef": "6ae9c7b3a469d23ffb7e8f6c",
      "ruleId": "R121",
      "ruleName": "unknown_sof",
      "ruleVersion": 3,
      "caseType": "Fraud",
      "riskScore": 85.0,
      "riskLabel": "High",
      "domain": "Entity Risk / KYC",
      "mainDomain": "Entity / Customer Risk Rule",
      "category": "dsl-natural",
      "engine": "predicate",
      "appliesTo": "transaction",
      "source": "mongo",
      "logicSource": "corpus",
      "alertable": true,
      "conditionSummary": "Source of funds = UNKNOWN",
      "explanation": "Flags entities with unknown or unverified source of funds.",
      "slaHours": null,
      "detector": "rule",
      "matchedConditions": [
        {
          "field": "customer.personalKyc.funds_wealth.source_of_funds",
          "resolvedField": "customer.personalKyc.funds_wealth.source_of_funds",
          "operator": "eq",
          "expected": "UNKNOWN",
          "actual": "UNKNOWN"
```

```
      }
    ]
  }
],
"ruleHitCount": 2,
"modelDetections": {
  "ml": {
    "detector": "ml",
    "modelVersion": "20261103T113836Z-mongo",
    "modelSource": "mongo",
    "algorithm": "lightgbm",
    "explainer": "TreeSHAP",
    "predictedClass": "High",
    "probabilities": {"Low": 0.000854, "Medium": 0.008331, "High": 0.990815},
    "threshold": 0.3,
    "riskScore": 99,
    "riskLabel": "High",
    "topFactors": [
      {
        "rank": 1,
        "feature": "source_of_funds",
        "fields": ["customer.personalKyc.funds_wealth.source_of_funds"],
        "group": "customer",
        "value": "UNKNOWN",
        "contribution": 5.647469,
        "direction": "increases_risk"
      },
      {
        "rank": 2,
        "feature": "sender_account_open_days",
        "fields": ["account.openedAt"],
        "group": "customer",
        "value": 2015.0,
        "contribution": 0.228111,
        "direction": "increases_risk"
      }
    ],
    "mitigatingFactors": [
      {
        "rank": 1,
        "feature": "velocity_7d_change",
        "fields": ["derived.velocity_7d_change"],
        "group": "collective",
        "value": 155.882353,
        "contribution": -0.373583,
        "direction": "decreases_risk"
      }
    ],
    "inferredCaseType": "Fraud",
    "inferredDomain": "Entity Risk / KYC",
    "inferenceBasis": "rule-field attribution of top model factors"
  },
  "dl": {
    "detector": "dl",
    "modelVersion": "20261103T113836Z-mongo",
```

```
      "modelSource": "mongo",
      "algorithm": "embedding-mlp",
      "explainer": "IntegratedGradients",
      "predictedClass": "High",
      "probabilities": {"Low": 1.7e-05, "Medium": 0.000213, "High": 0.999771},
      "threshold": 0.44,
      "riskScore": 100,
      "riskLabel": "High",
      "topFactors": [
        {
          "rank": 1,
          "feature": "aml_status",
          "fields": ["customer.amlStatus"],
          "group": "customer",
          "value": "clear",
          "contribution": 0.725375,
          "direction": "increases_risk"
        }
      ],
      "mitigatingFactors": [
        {
          "rank": 1,
          "feature": "citizenship_country",
          "fields": ["customer.country"],
          "group": "customer",
          "value": "Bangladesh",
          "contribution": -1.045994,
          "direction": "decreases_risk"
        }
      ],
      "inferredCaseType": "Fraud",
      "inferredDomain": "Entity Risk / Industry Risk",
      "inferenceBasis": "rule-field attribution of top model factors"
    }
  },
  "triggeredFields": [
    {
      "field": "customer.personalKyc.funds_wealth.source_of_funds",
      "weight": 1.696958,
      "detectors": ["rule", "ml"],
      "sources": [
        {"detector": "rule", "ref": "R121", "how": "matched_condition", "weight":
0.85},
        {"detector": "ml", "ref": "20261103T113836Z-mongo", "how":
"feature:source_of_funds", "weight": 0.846958}
      ],
      "value": "UNKNOWN",
      "detectorCount": 2
    },
    {
      "field": "derived.amount_pct_decrease_prev",
      "weight": 0.65,
      "detectors": ["rule"],
      "sources": [
        {"detector": "rule", "ref": "R83", "how": "matched_condition", "weight":
```

```
0.65}
      ],
      "value": 54.738907,
      "detectorCount": 1
    }
  ],
  "topTriggeredFields": [
    "customer.personalKyc.funds_wealth.source_of_funds",
    "derived.amount_pct_decrease_prev",
    "customer.amlStatus"
  ],
  "reasons": [
    {
      "detector": "rule",
      "ref": "R121",
      "text": "Rule R121 unknown_sof [Fraud/High]:
customer.personalKyc.funds_wealth.source_of_funds eq UNKNOWN (observed UNKNOWN)"
    },
    {
      "detector": "rule",
      "ref": "R83",
      "text": "Rule R83 sudden_amt_drop [Fraud/Medium]: Transaction amount decrease
gt 50% vs previous (observed 54.738907)"
    }
  ],
  "categories": {
    "caseTypes": ["Fraud"],
    "domains": ["Entity Risk / KYC", "Velocity & Pattern Shift"],
    "mainDomains": ["Entity / Customer Risk Rule"]
  },
  "collectiveContext": {
    "senderTxCount24h": 3.0,
    "senderSum24h": 107527.2,
    "senderSumLast5": 353154.22,
    "senderUniqueBeneficiariesDay": 2,
    "beneficiaryTxCount24h": 1.0,
    "beneficiaryUniqueSendersDay": 1,
    "secsSincePrevSenderTx": 37440.0,
    "amountPctChangePrev": -54.738907,
    "senderAmountZscore30d": -0.366413,
    "senderTxCount30d": 34.0,
    "txFreqChangePct": 155.882353,
    "structuringCount24h": 0.0,
    "passThroughRatio": 0.0
  },
  "pipeline": {
    "modelVersions": {"ml": "20261103T113836Z-mongo", "dl":
"20261103T113836Z-mongo"},
    "appVersion": "1.0.0",
    "rulesEvaluated": 149
  },
  "alert": {"alertId": "6ae9c802a469d23ffb7ec03a", "uid": "AL-0000102", "status":
"new", "action": "created"},
  "case": {"caseId": "6ae9c802a469d23ffb7ec0a4", "uid": "CA-1793705986411-234114",
"action": "created"},
```

```
  "updatedAt": "2026-11-03T11:39:46.082000Z",
  "createdAt": "2026-11-03T11:39:46.082000Z",
  "runHistory": [
    {
      "runId": "MON-20261102T180000Z-828c0c",
      "detectedAt": "2026-11-03T11:39:46.082000Z",
      "riskScore": 100,
      "detectedBy": ["rule", "ml", "dl"]
    }
  ]
}
```


GET /results/timeseries?start=2026-11-01&end=2026-11-03

The response has one entry per day that has detections; days with none are left out.


```
{
  "start": "2026-11-01",
  "end": "2026-11-03",
  "dateField": "run",
  "series": [
    {
      "date": "2026-11-03",
      "total": 106,
      "avgScore": 97.86,
      "byDetector": {"dl": 89, "ml": 96, "rule": 102},
      "byRiskLabel": {"Critical": 1, "High": 101, "Medium": 4},
      "byCaseType": {"AML": 4, "Fraud": 102}
    }
  ]
}
```


GET /runs?jobType=monitoring&limit=1


```
[
  {
    "runId": "MON-20261102T180000Z-828c0c",
    "jobType": "monitoring",
    "status": "succeeded",
    "startedAt": "2026-11-03T11:39:40.515000Z",
    "window": {"start": "2026-11-01T18:00:00Z", "end": "2026-11-02T18:00:00Z"},
    "timezone": "Asia/Dhaka",
    "dryRun": false,
    "runDate": "2026-11-03",
    "warnings": [],
    "counts": {
      "transactions": 330,
      "historyRows": 9370,
      "risky": 106,
      "byDetector": {"rule": 102, "ml": 96, "dl": 89},
      "byDetectorCombination": {"rule+ml": 16, "rule+ml+dl": 80, "dl": 4, "rule+dl":
5, "rule": 1},
      "byRiskLabel": {"High": 101, "Medium": 4, "Critical": 1},
      "byCaseType": {"Fraud": 102, "AML": 4},
      "topFields": [
```

```
        {"field": "amount", "count": 45},
        {"field": "derived.velocity_7d_change", "count": 42}
      ],
      "alerts": {"created": 106},
      "cases": {"casesCreated": 65, "casesUpdated": 0, "alertsLinked": 106,
"skippedReason": null}
    },
    "modelVersions": {"ml": "20261103T113836Z-mongo", "dl":
"20261103T113836Z-mongo"},
    "ruleSummary": {
      "rulesTotal": 597,
      "byStatus": {"evaluated": 144, "unresolvable": 91, "skipped_manual": 212,
"skipped_customer_scope": 117},
      "rulesFired": 37,
      "topRules": [{"ruleId": "R83", "count": 124}, {"ruleId": "R82", "count": 121}],
      "transactionsFlagged": 269,
      "rowsEvaluated": 330
    },
    "finishedAt": "2026-11-03T11:39:50.117000Z",
    "durationSec": 9.6
  }
]
```


`ruleSummary.transactionsFlagged` counts every rule hit, including Medium-only hits. `counts.risky` counts only transactions that passed the risk threshold.

GET /models?modelType=ml&limit=1


```
[
  {
    "modelType": "ml",
    "version": "20261103T113836Z-mongo",
    "path": "artifacts/ml/20261103T113836Z-mongo",
    "status": "active",
    "registeredAt": "2026-11-03T11:38:45.230000Z",
    "source": "mongo",
    "highThreshold": 0.3,
    "metrics": {"val_macro_f1": 0.9848, "test_macro_f1": 0.9805, "test_high_pr_auc":
0.9972},
    "data": {
      "dataset": "mongo",
      "window": {"start": "2026-10-03T18:00:00Z", "end": "2026-11-02T18:00:00Z"},
      "rowsUsed": 9394,
      "split": {"method": "time", "train": 6575, "val": 1409, "test": 1410},
      "label": {
        "scheme": "max fired rule riskLabel, overridden by analyst outcomes",
        "sources": {"alert_false_positive": 12, "rules": 9381, "alert_escalated": 1}
      },
      "classDistribution": {"Low": 1977, "Medium": 5259, "High": 2158}
    },
    "statusReason": "forced",
    "statusAt": "2026-11-03T11:38:45.231000Z"
  }
]
```

GET /rules/coverage

The response shows one example rule per status. Filter with `?status=unresolvable` to list the rules that can never fire on current data.


```
{
  "runId": "MON-20261102T180000Z-828c0c",
  "startedAt": "2026-11-03T11:39:40.515000Z",
  "byStatus": {
    "evaluated": 144, "unresolvable": 91, "skipped_manual": 212,
"skipped_customer_scope": 117,
    "partially_resolvable": 5, "skipped_screening": 16, "skipped_no_logic": 12
  },
  "rules": [
    {
      "ruleId": "R231",
      "ruleName": "ben_medium_high_risk",
      "engine": "predicate",
      "appliesTo": "transaction",
      "logicSource": "corpus",
      "source": "mongo",
      "firedCount": 0,
      "issues": [{"field": "Beneficiary risk tier", "issue": "unresolved_field"}],
      "status": "unresolvable"
    },
    {
      "ruleId": "R161",
      "ruleName": "missing_gdpr_basis",
      "engine": "manual",
      "appliesTo": "customer",
      "logicSource": "corpus",
      "source": "mongo",
      "firedCount": 0,
      "issues": [],
      "status": "skipped_manual"
    }
  ]
}
```


POST /runs/monitoring, /health and errors

Call	Status	Body
`POST /runs/monitoring` with `{"dryRun": true}`	202	`{"accepted": true, "runId": "MON-API-2647866c1f"}`
`GET /health`	200	`{"status": "ok", "database": true, "version": "1.0.0"}`
No or wrong `X-API-Key`	401	`{"detail": "invalid or missing X-API-Key"}`
Unknown transaction	404	`{"detail": "no AI monitoring result for this transaction"}`

Call	Status	Body
`date=03-11-2026`	422	`{"detail": "date must be YYYY-MM-DD"}`

The `runId` from a POST can be passed to `GET /runs/{runId}` to follow the run.

Examples

```
BASE=http://[TM_HOST]:5054/api/v1/tm
KEY="X-API-Key: [YOUR_API_KEY]"

# Yesterday's risky transactions, highest score first, with summary
curl -H "$KEY" "$BASE/results/daily?date=2026-11-03&limit=50"

# High-risk AML detections by the ML model in October, by transaction date
curl -H "$KEY"
"$BASE/results?start=2026-10-01&end=2026-10-31&detector=ml&caseType=AML&riskLabel=Hig
h&dateField=transaction"

# Daily trend for a chart
curl -H "$KEY" "$BASE/results/timeseries?start=2026-10-01&end=2026-11-03"

# Full evidence for one transaction
curl -H "$KEY" "$BASE/results/6ae8914b01ba00732f3ca9df"
```

Things to know when integrating:

- New results appear after the 01:00 run. Poll once a day, or check `/runs` for the latest successful run.
- The run date is the day after the activity. To see when transactions actually happened, use `dateField=transaction`.
- Alerts and cases live in the shared `alerts` and `cases` collections. Use the `alert` and `case` IDs in each result to link to them.